Screen Capture

Adversaries may attempt to take screen captures of the desktop to gather information over the course of an operation. Screen capturing functionality may be included as a feature of a remote access tool used in post-compromise operations. Taking a screenshot is also typically possible through native utilities or API calls, such as `CopyFromScreen`, `xwd`, or `screencapture`.[1][2]

ID: T1113

Sub-techniques: No sub-techniques

ⓘ Tactic: Collection

ⓘ Platforms: Linux, Windows, macOS

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Version Permalink

Procedure Examples

ID	Name	Description
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks, the adversaries captured screenshots of devices using `nircmd` console through the command `nircmd.exe "savescreenshot C:\Windows\Temp\imagetmp.png`.[3]
S0331	Agent Tesla	Agent Tesla can capture screenshots of the victim's desktop.[4][5][6][7][8]
S0622	AppleSeed	AppleSeed can take screenshots on a compromised host by calling a series of APIs.[9][10]
G0007	APT28	APT28 has used tools to take screenshots from victims.[11][12][13][14]
G0087	APT39	APT39 has used a screen capture utility to take screenshots on a compromised host.[15][16]
G1044	APT42	APT42 has used malware, such as GHAMBAR and POWERPOST, to take screenshots.[17]
S0456	Aria-body	Aria-body has the ability to capture screenshots on compromised hosts.[18]
S9031	AshTag	The AshTag AshenOrchestrator component has the ability to take screenshots.[19]
S1087	AsyncRAT	AsyncRAT has the ability to view the screen on compromised hosts.[20]
S0438	Attor	Attor's has a plugin that captures screenshots of the target applications.[21]
S0344	Azorult	Azorult can capture screenshots of the victim's machines.[22]
S1081	BADHATCH	BADHATCH can take screenshots and send them to an actor-controlled C2 server.[23]
S0128	BADNEWS	BADNEWS has a command to take a screenshot and send it to the C2 server.[24][25]
S0337	BadPatch	BadPatch captures screenshots in .jpg format and then exfiltrates them.[26]
S0234	Bandook	Bandook is capable of taking an image of and uploading the current desktop.[27][28]
S0017	BISCUIT	BISCUIT has a command to periodically take screenshots of the system.[29]
S0089	BlackEnergy	BlackEnergy is capable of taking screenshots.[30]
S0657	BLUELIGHT	BLUELIGHT has captured a screenshot of the display every 30 seconds for the first 5 minutes after initiating a C2 loop, and then once every five minutes thereafter.[31]
G0060	BRONZE BUTLER	BRONZE BUTLER has used a tool to capture screenshots.[32][33]
S1063	Brute Ratel C4	Brute Ratel C4 can take screenshots on compromised hosts.[34]
S0454	Cadelspy	Cadelspy has the ability to capture screenshots and webcam photos.[35]
S0351	Cannon	Cannon can take a screenshot of the desktop.[36]
S0030	Carbanak	Carbanak performs desktop video recording and captures screenshots of the desktop and sends it to the C2 server.[37]
S0484	Carberp	Carberp can capture display screenshots with the screens_dll.dll plugin.[38]
S0348	Cardinal RAT	Cardinal RAT can capture screenshots.[39]

ID	Name	Description
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can capture screenshots on targeted systems using a timer and either upload them or store them to disk.[43]
S0023	CHOPSTICK	CHOPSTICK has the capability to capture screenshots.[13]
S0667	Chrommme	Chrommme has the ability to capture screenshots.[44]
S0660	Clambling	Clambling has the ability to capture screenshots.[45]
S0154	Cobalt Strike	Cobalt Strike's Beacon payload is capable of capturing screenshots.[46][47][48]
S0338	Cobian RAT	Cobian RAT has a feature to perform screen capture.[49]
S0591	ConnectWise	ConnectWise can take screenshots on remote hosts.[50]
S0050	CosmicDuke	CosmicDuke takes periodic screenshots and exfiltrates them.[51]
S0115	Crimson	Crimson contains a command to perform screen captures.[52][53][54]
S0235	CrossRAT	CrossRAT is capable of taking screen captures.[27]
S1153	Cuckoo Stealer	Cuckoo Stealer can run `screencapture` to collect screenshots from compromised hosts. [55]
G0070	Dark Caracal	Dark Caracal took screenshots using their Windows malware.[27]
S0187	Daserf	Daserf can take screenshots.[56][32]
S0021	Derusbi	Derusbi is capable of performing screen captures.[57]
S0213	DOGCALL	DOGCALL is capable of capturing screenshots of the victim's machine.[58][59]
G0035	Dragonfly	Dragonfly has performed screen captures of victims, including by using a tool, scr.exe (which matched the hash of ScreenUtil).[60][61][62]
S1159	DUSTTRAP	DUSTTRAP can capture screenshots.[63]
S0062	DustySky	DustySky captures PNG screenshots of the main screen.[64]
S0593	ECCENTRICBANDWAGON	ECCENTRICBANDWAGON can capture screenshots and store them locally.[65]
S0363	Empire	Empire is capable of capturing screenshots on Windows and macOS systems.[66]
S0152	EvilGrab	EvilGrab has the capability to capture screenshots.[67]
G0046	FIN7	FIN7 captured screenshots and desktop video recordings.[68]
S0182	FinFisher	FinFisher takes a screenshot of the screen and displays it on top of all other windows for few seconds in an apparent attempt to hide some messages showed by the system during the setup process.[69][70]
S0143	Flame	Flame can take regular screenshots when certain applications are open that are sent to the command and control server.[71]
S0381	FlawedAmmyy	FlawedAmmyy can capture screenshots.[72]

ID	Name	Description
G0115	GOLD SOUTHFIELD	GOLD SOUTHFIELD has used the remote monitoring and management tool ConnectWise to obtain screen captures from victim's machines.[79]
S0417	GRIFFON	GRIFFON has used a screenshot module that can be used to take a screenshot of the remote system.[80]
G0043	Group5	Malware used by Group5 is capable of watching the victim's screen.[81]
S0151	HALFBAKED	HALFBAKED can obtain screenshots from the victim.[82]
S1229	Havoc	Havoc can capture screenshots.[83][84][85]
S0431	HotCroissant	HotCroissant has the ability to do real time screen viewing on an infected host.[86]
S9007	HTTPTroy	HTTPTroy has obtained screen captures leveraging the `screen` command which captures, encrypts and uploads the stolen image to the adversary controlled C2 server.[87]
S0203	Hydraq	Hydraq includes a component based on the code of VNC that can stream a live feed of the desktop of an infected host.[88]
S0398	HyperBro	HyperBro has the ability to take screenshots.[89]
S0260	InvisiMole	InvisiMole can capture screenshots of not only the entire screen, but of each separate window open, in case they are overlapping.[90][91]
S0163	Janicab	Janicab captured screenshots and sent them out to a C2 server.[92][93]
S0044	JHUHUGIT	A JHUHUGIT variant takes screenshots by simulating the user pressing the "Take Screenshot" key (VK_SCREENSHOT), accessing the screenshot saved in the clipboard, and converting it to a JPG image.[94][95]
S0283	jRAT	jRAT has the capability to take screenshots of the victim's machine.[96][97]
S0088	Kasidet	Kasidet has the ability to initiate keylogging and screen captures.[98]
S0265	Kazuar	Kazuar captures screenshots of the victim's screen.[99]
S0387	KeyBoy	KeyBoy has a command to perform screen grabbing.[100]
S0271	KEYMARBLE	KEYMARBLE can capture screenshots of the victim's machine.[101]
G0094	Kimsuky	Kimsuky has captured browser screenshots using TRANSLATEXT.[102] Kimsuky has also obtained screen captures with custom malware.[87]
S0437	Kivars	Kivars has the ability to capture screenshots on the infected host.[103]
S0356	KONNI	KONNI can take screenshots of the victim's machine.[104]
S1185	LightSpy	LightSpy uses Apple's built-in AVFoundation Framework library to access the user's camera and screen. It uses the `AVCaptureStillImage` to take a picture using the user's camera and the `AVCaptureScreen` to take a screenshot or record the user's screen for a specified period of time.[105]
S0680	LitePower	LitePower can take system screenshots and save them to `%AppData%`.[106]
S0681	Lizar	Lizar can take JPEG screenshots of an infected system.[107][108] Lizar has also used a plugin to take a

ID	Name	Description
S1142	LunarMail	LunarMail can capture screenshots from compromised hosts.[114]
S0409	Machete	Machete captures screenshots.[115][116][117][118]
S1016	MacMa	MacMa has used Apple's Core Graphic APIs, such as `CGWindowListCreateImageFromArray`, to capture the user's screen and open windows.[119][120]
S0282	MacSpy	MacSpy can capture screenshots of the desktop over multiple monitors.[73]
S1060	Mafalda	Mafalda can take a screenshot of the target machine and save it to a file.[121]
G0059	Magic Hound	Magic Hound malware can take a screenshot and upload the file to its C2 server.[122]
S1156	Manjusaka	Manjusaka can take screenshots of the victim desktop.[123]
S0652	MarkiRAT	MarkiRAT can capture screenshots that are initially saved as 'scr.jpg'.[124]
S0167	Matryoshka	Matryoshka is capable of performing screen captures.[125][126]
S1059	metaMain	metaMain can take and save screenshots.[121][127]
S0455	Metamorfo	Metamorfo can collect screenshots of the victim's machine.[128][129]
S0339	Micropsia	Micropsia takes screenshots every 90 seconds by calling the Gdi32.BitBlt API.[130]
S1122	Mispadu	Mispadu has the ability to capture screenshots on compromised hosts.[131][132][133][134]
G1019	MoustachedBouncer	MoustachedBouncer has used plugins to take screenshots on targeted systems.[135]
G0069	MuddyWater	MuddyWater has used malware that can capture screenshots of the victim's machine.[136]
S0198	NETWIRE	NETWIRE can capture the victim's screen.[137][138][139][140]
S1090	NightClub	NightClub can load a module to call `CreateCompatibleDC` and `GdipSaveImageToStream` for screen capture.[135]
S0385	njRAT	njRAT can capture screenshots of the victim's machines.[141][142]
S1107	NKAbuse	NKAbuse can take screenshots of the victim machine.[143]
S0644	ObliqueRAT	ObliqueRAT can capture a screenshot of the current screen.[144]
S0340	Octopus	Octopus can capture screenshots of the victims' machine.[145][146][147]
G0049	OilRig	OilRig has a tool called CANDYKING to capture a screenshot of user's desktop.[148]
S1050	PcShare	PcShare can take screen shots of a compromised machine.[74]
S0643	Peppy	Peppy can take screenshots on targeted systems.[52]
S0013	PlugX	PlugX allows the operator to capture screenshots.[149]
S0428	PoetRAT	PoetRAT has the ability to take screen captures.[150][151]

ID	Name	Description
S0113	Prikormka	Prikormka contains a module that captures screenshots of the victim's desktop.[157]
S0279	Proton	Proton captures the content of the desktop with the screencapture binary.[73]
S0147	Pteranodon	Pteranodon can capture screenshots at a configurable interval.[158][159]
S0192	Pupy	Pupy can drop a mouse-logger that will take small screenshots around at each click and then send back to the server.[160]
S1209	Quick Assist	Quick Assist allows for the remote administrator to take screenshots of the running system.[161]
S0686	QuietSieve	QuietSieve has taken screenshots every five minutes and saved them to the user's local Application Data folder under `Temp\SymbolSourceSymbols\icons` or `Temp\ModeAuto\icons`.[162]
S1148	Raccoon Stealer	Raccoon Stealer can capture screenshots from victim systems.[163][164]
S0629	RainyDay	RainyDay has the ability to capture screenshots.[165]
S0458	Ramsay	Ramsay can take screenshots every 30 seconds as well as when an external removable storage device is connected.[166]
S0662	RCSession	RCSession can capture screenshots from a compromised host.[167]
S0495	RDAT	RDAT can take a screenshot on the infected system.[168]
S0153	RedLeaves	RedLeaves can capture screenshots.[169][170]
S1240	RedLine Stealer	RedLine Stealer can capture screenshots on a compromised host.[171][172]
S0332	Remcos	Remcos takes automated screenshots of the infected machine.[173][174]
S0375	Remexi	Remexi takes screenshots of windows of interest.[175]
S0592	RemoteUtilities	RemoteUtilities can take screenshots on a compromised host.[176]
S0379	Revenge RAT	Revenge RAT has a plugin for screen capture.[177]
S0270	RogueRobin	RogueRobin has a command named `$screenshot` that may be responsible for taking screenshots of the victim machine.[178]
S0240	ROKRAT	ROKRAT can capture screenshots of the infected system using the `gdi32` library.[179][180][181][182][183]
S0090	Rover	Rover takes screenshots of the compromised system's desktop and saves them to `C:\system\screenshot.bmp` for exfiltration every 60 minutes.[184]
S0148	RTM	RTM can capture screenshots.[185][186]
S0546	SharpStage	SharpStage has the ability to capture the victim's screen.[187][188]
S0217	SHUTTERSPEED	SHUTTERSPEED can capture screenshots.[58]
G0091	Silence	Silence can capture victim screen activity.[189][190]
S0692	SILENTTRINITY	SILENTTRINITY can take a screenshot of the current desktop.[191]

ID	Name	Description
S0273	Socksbot	Socksbot can take screenshots.[196]
S0380	StoneDrill	StoneDrill can take screenshots.[197]
S1034	StrifeWater	StrifeWater has the ability to take screen captures.[198]
S1064	SVCReady	SVCReady can take a screenshot from an infected host.[199]
S0663	SysUpdate	SysUpdate has the ability to capture screenshots.[200]
S0098	T9000	T9000 can take screenshots of the desktop and target application windows, saving them to user directories as one byte XOR encrypted .dat files.[201]
S0467	TajMahal	TajMahal has the ability to take screenshots on an infected host including capturing content from windows of instant messaging applications.[202]
S0004	TinyZBot	TinyZBot contains screen capture functionality.[203]
S1239	TONESHELL	TONESHELL has conducted screen capturing.[204]
S1201	TRANSLATEXT	TRANSLATEXT has the ability to capture screenshots of new browser tabs, based on the presence of the `Capture` flag.[102]
S0094	Trojan.Karagany	Trojan.Karagany can take a desktop screenshot and save the file into `\ProgramData\Mail\MailAg\shot.png`.[205][206]
S1196	Troll Stealer	Troll Stealer can capture screenshots from victim machines.[207][208]
S0647	Turian	Turian has the ability to take screenshots.[209]
S0199	TURNEDUP	TURNEDUP is capable of taking screenshots.[210]
S0275	UPPERCUT	UPPERCUT can capture desktop screenshots in the PNG format and send them to the C2 server.[211][212][213][214]
S0386	Ursnif	Ursnif has used hooked APIs to take screenshots.[215][216]
S0476	Valak	Valak has the ability to take screenshots on a compromised host.[217]
S0257	VERMIN	VERMIN can perform screen captures of the victim’s machine.[218]
G1055	VOID MANTICORE	VOID MANTICORE has captured screen content during an active Zoom session.[219]
G1017	Volt Typhoon	Volt Typhoon has obtained a screenshot of the victim's system using the gdi32.dll and gdiplus.dll libraries.[220]
G1035	Winter Vivern	Winter Vivern delivered PowerShell scripts capable of taking screenshots of victim machines.[221]
S1065	Woody RAT	Woody RAT has the ability to take a screenshot of the infected host desktop using Windows GDI+.[222]
S0161	XAgentOSX	XAgentOSX contains the takeScreenShot (along with startTakeScreenShot and stopTakeScreenShot) functions to take screenshots using the CGGetActiveDisplayList, CGDisplayCreateImage, and NSImage:initWithCGImage methods.[12]

ID	Name	Description
S0251	Zebrocy	A variant of Zebrocy captures screenshots of the victim's machine in JPEG and BMP format.[36][227][228][229][230][231]
S0330	Zeus Panda	Zeus Panda can take screenshots of the victim's machine.[232]
S0086	ZLib	ZLib has the ability to obtain screenshots of the compromised system.[233]
S0412	ZxShell	ZxShell can capture screenshots.[234]

Mitigations

This type of attack technique cannot be easily mitigated with preventive controls since it is based on the abuse of system features.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0346	Detect Screen Capture via Commands and API Calls	AN0980	Unusual use of screen capture APIs (e.g., CopyFromScreen) or command-line tools to write image files to disk.
		AN0981	Invocation of built-in commands like screencapture or use of undocumented APIs from suspicious parent processes.
		AN0982	Use of tools like xwd or import to generate screenshots, especially under non-GUI parent processes.

References

1. Microsoft. (n.d.). Graphics.CopyFromScreen Method. Retrieved March 24, 2020.
2. Thomas Reed. (2017, January 18). New Mac backdoor using antiquated code. Retrieved July 5, 2017.
3. CERT Polska. (2026, January 30). Energy Sector Incident Report – 29 December. Retrieved April 22, 2026.
4. Brumaghin, E., et al. (2018, October 15). Old dog, new tricks - Analysing new RTF-based campaign distributing Agent Tesla, Loki with PyREbox. Retrieved November 5, 2018.
5. The DigiTrust Group. (2017, January 12). The Rise of Agent Tesla. Retrieved November 5, 2018.
6. Zhang, X. (2018, April 05). Analysis of New Agent Tesla Spyware Variant. Retrieved November 5, 2018.
7. Zhang, X. (2017, June 28). In-Depth Analysis of A New Variant of .NET Malware AgentTesla. Retrieved November 5, 2018.
8. Arsene, L. (2020, April 21). Oil & Gas Spearphishing Campaigns Drop Agent Tesla Spyware in Advance of Historic OPEC+ Deal. Retrieved May 19, 2020.
9. Jazi, H. (2021, June 1). Kimsuky APT continues to target South Korean government using AppleSeed backdoor. Retrieved June 10, 2021.
10. KISA. (2021). Phishing Target Reconnaissance and Attack Resource Analysis Operation Muzabi. Retrieved March 8, 2024.
11. ESET. (2016, October). En Route with Sednit - Part 2: Observing the Comings and Goings. Retrieved November 21, 2016.
12. Robert Falcone. (2017, February 14). XAgentOSX: Sofacy's Xagent macOS Tool. Retrieved July 12, 2017.
13. Mueller, R. (2018, July 13). Indictment - United States of America vs. VIKTOR BORISOVICH NETYKSHO, et al. Retrieved November 17, 2024.
14. Secureworks CTU. (2017, March 30). IRON TWILIGHT Supports Active Measures. Retrieved February 28, 2022.
15. Symantec. (2018, February 28). Chafer: Latest Attacks Reveal Heightened Ambitions. Retrieved May 22, 2020.
16. FBI. (2020, September 17). Indicators of Compromise Associated with Rana Intelligence Computing, also known as Advanced Persistent Threat 39, Chafer, Cadelspy, Remexi, and ITG07. Retrieved December 10, 2020.
17. Mandiant. (n.d.). APT42: Crooked Charms, Cons and Compromises. Retrieved October 9, 2024.
18. CheckPoint. (2020, May 7). Naikon APT: Cyber Espionage Reloaded. Retrieved May 26, 2020.
19. Unit 42. (2025, December 11). Hamas-Affiliated Ashen Lepus Targets Middle Eastern Diplomatic Entities With New AshTag Malware Suite. Retrieved April 20, 2026.
20. Nyan-x-Cat. (n.d.). NYAN-x-CAT / AsyncRAT-C-Sharp. Retrieved October 3, 2023.
21. Hromcova, Z. (2019, October). AT COMMANDS, TOR-BASED COMMUNICATIONS: MEET ATTOR, A FANTASY CREATURE AND ALSO A SPY PLATFORM. Retrieved May 6, 2020.
22. Yan, T., et al. (2018, November 21). New Wine in Old Bottle: New Azorult Variant Found in FindMyName Campaign using Fallout Exploit Kit. Retrieved November 29, 2018.
23. Vrabie, V., et al. (2021, March 10). FIN8 Returns with Improved BADHATCH Toolkit. Retrieved September 8, 2021.
118. kate. (2020, September 25). APT-C-43 steals Venezuelan military secrets to provide intelligence support for the reactionaries — HpReact campaign. Retrieved November 20, 2020.
119. M.Léveillé, M., Cherepanov, A.. (2022, January 25). Watering hole deploys new macOS malware, DazzleSpy, in Asia. Retrieved May 6, 2022.
120. Wardle, P. (2021, November 11). OSX.CDDS (OSX.MacMa). Retrieved June 30, 2022.
121. Ehrlich, A., et al. (2022, September). THE MYSTERY OF METADOR | AN UNATTRIBUTED THREAT HIDING IN TELCOS, ISPS, AND UNIVERSITIES. Retrieved January 23, 2023.
122. Lee, B. and Falcone, R. (2017, February 15). Magic Hound Campaign Attacks Saudi Targets. Retrieved December 27, 2017.
123. Asheer Malhotra & Vitor Ventura. (2022, August 2). Manjusaka: A Chinese sibling of Sliver and Cobalt Strike. Retrieved September 4, 2024.
124. GReAT. (2021, June 16). Ferocious Kitten: 6 Years of Covert Surveillance in Iran. Retrieved September 22, 2021.
125. ClearSky Cyber Security and Trend Micro. (2017, July). Operation Wilted Tulip: Exposing a cyber espionage apparatus. Retrieved August 21, 2017.
126. Minerva Labs LTD and ClearSky Cyber Security. (2015, November 23). CopyKittens Attack Group. Retrieved November 17, 2024.
127. SentinelLabs. (2022, September 22). Metador Technical Appendix. Retrieved April 4, 2023.
128. Sierra, E., Iglesias, G.. (2018, April 24). Metamorfo Campaigns Targeting Brazilian Users. Retrieved July 30, 2020.
129. ESET Research. (2019, October 3). Casbaneiro: peculiarities of this banking Trojan that affects Brazil and Mexico. Retrieved September 23, 2021.
130. Tsarfaty, Y. (2018, July 25). Micropsia Malware. Retrieved November 13, 2018.
131. SCILabs. (2021, December 23). Cyber Threat Profile Malteiro. Retrieved March 13, 2024.
132. SCILabs. (2023, May 23). Evolution of banking trojan URSA/Mispadu. Retrieved March 13, 2024.
133. ESET Security. (2019, November 19). Mispadu: Advertisement for a discounted Unhappy Meal. Retrieved March 13, 2024.
134. Garcia, F., Regalado, D. (2023, March 7). Inside Mispadu massive infection campaign in LATAM. Retrieved March 15, 2024.
135. Faou, M. (2023, August 10). MoustachedBouncer: Espionage against foreign diplomats in Belarus. Retrieved September 25, 2023.
136. Kaspersky Lab's Global Research & Analysis Team. (2018, October 10). MuddyWater expands operations. Retrieved November 2, 2018.
137. McAfee. (2015, March 2). Netwire RAT Behind Recent Targeted Attacks. Retrieved February 15, 2018.
138. Maniath, S. and Kadam P. (2019, March 19). Dissecting a NETWIRE Phishing Campaign's Usage of Process Hollowing. Retrieved January 7, 2021.
139. Lambert, T. (2020, January 29). Intro to Netwire. Retrieved January 7, 2021.
140. Proofpoint. (2020, December 2). Geofenced NetWire Campaigns. Retrieved January 7, 2021.

28. Check Point. (2020, November 26). Bandook: Signed & Delivered. Retrieved May 31, 2021.
29. Mandiant. (n.d.). Appendix C (Digital) - The Malware Arsenal. Retrieved July 18, 2016.
30. Baumgartner, K. and Garnaeva, M.. (2014, November 3). BE2 custom plugins, router abuse, and target profiles. Retrieved March 24, 2016.
31. Cash, D., Grunzweig, J., Meltzer, M., Adair, S., Lancaster, T. (2021, August 17). North Korean APT InkySquid Infects Victims Using Browser Exploits. Retrieved September 30, 2021.
32. Counter Threat Unit Research Team. (2017, October 12). BRONZE BUTLER Targets Japanese Enterprises. Retrieved January 4, 2018.
33. Chen, J. et al. (2019, November). Operation ENDTRADE: TICK's Multi-Stage Backdoors for Attacking Industries and Stealing Classified Data. Retrieved June 9, 2020.
34. Harbison, M. and Renals, P. (2022, July 5). When Pentest Tools Go Brutal: Red-Teaming Tool Being Abused by Malicious Actors. Retrieved February 1, 2023.
35. Symantec Security Response. (2015, December 7). Iran-based attackers use back door threats to spy on Middle Eastern targets. Retrieved April 17, 2019.
36. Falcone, R., Lee, B. (2018, November 20). Sofacy Continues Global Attacks and Wheels Out New 'Cannon' Trojan. Retrieved November 26, 2018.
37. Bennett, J., Vengerik, B. (2017, June 12). Behind the CARBANAK Backdoor. Retrieved June 11, 2018.
38. Giuliani, M., Allievi, A. (2011, February 28). Carberp - a modular information stealing trojan. Retrieved September 12, 2024.
39. Grunzweig, J.. (2017, April 20). Cardinal RAT Active for Over Two Years. Retrieved December 8, 2018.
40. Balanza, M. (2018, April 02). Infostealer.Catchamas. Retrieved November 17, 2024.
41. Salem, E. (2020, November 17). CHAES: Novel Malware Targeting Latin American E-Commerce. Retrieved June 30, 2021.
42. Check Point. (2022, January 11). APT35 exploits Log4j vulnerability to distribute new modular PowerShell toolkit. Retrieved January 24, 2022.
43. Jenkins, L. at al. (2022, August 4). ROADSWEEP Ransomware - Likely Iranian Threat Actor Conducts Politically Motivated Disruptive Activity Against Albanian Government Organizations. Retrieved August 6, 2024.
44. Dupuy, T. and Faou, M. (2021, June). Gelsemium. Retrieved November 30, 2021.
45. Lunghi, D. et al. (2020, February). Uncovering DRBControl. Retrieved November 12, 2021.
46. Strategic Cyber LLC. (2017, March 14). Cobalt Strike Manual. Retrieved May 24, 2017.
47. Amnesty International. (2021, February 24). Vietnamese activists targeted by notorious hacking group. Retrieved March 1, 2021.
48. Strategic Cyber LLC. (2020, November 5). Cobalt Strike: Advanced Threat Tactics for Penetration Testers. Retrieved April 13, 2021.
49. Yadav, A., et al. (2017, August 31). Cobian RAT – A backdoored RAT. Retrieved November 13, 2018.
50. Mele, G. et al. (2021, February 10). Probable Iranian Cyber Actors, Static Kitten, Conducting Cyberespionage Campaign Targeting UAE and Kuwait Government Agencies. Retrieved March 17, 2021.
51. F-Secure Labs. (2014, July). COSMICDUKE Cosmu with a twist of MiniDuke. Retrieved July 3, 2014.
144. Malhotra, A. (2021, March 2). ObliqueRAT returns with new campaign using hijacked websites. Retrieved September 2, 2021.
145. Kaspersky Lab's Global Research & Analysis Team. (2018, October 15). Octopus-infested seas of Central Asia. Retrieved November 14, 2018.
146. Paganini, P. (2018, October 16). Russia-linked APT group DustSquad targets diplomatic entities in Central Asia. Retrieved August 24, 2021.
147. Cherepanov, A. (2018, October 4). Nomadic Octopus Cyber espionage in Central Asia. Retrieved October 13, 2021.
148. Davis, S. and Caban, D. (2017, December 19). APT34 - New Targeted Attack in the Middle East. Retrieved December 20, 2017.
149. Computer Incident Response Center Luxembourg. (2013, March 29). Analysis of a PlugX variant. Retrieved November 5, 2018.
150. Mercer, W, et al. (2020, April 16). PoetRAT: Python RAT uses COVID-19 lures to target Azerbaijan public and private sectors. Retrieved April 27, 2020.
151. Dragos. (n.d.). ICS Cybersecurity Year in Review 2020. Retrieved February 25, 2021.
152. PowerShellMafia. (2012, May 26). PowerSploit - A PowerShell Post-Exploitation Framework. Retrieved February 6, 2018.
153. PowerSploit. (n.d.). PowerSploit. Retrieved February 6, 2018.
154. Singh, S. et al.. (2018, March 13). Iranian Threat Group Updates Tactics, Techniques and Procedures in Spear Phishing Campaign. Retrieved April 11, 2018.
155. Lunghi, D. and Horejsi, J.. (2019, June 10). MuddyWater Resurfaces, Uses Multi-Stage Backdoor POWERSTATS V3 and New Post-Exploitation Tools. Retrieved May 14, 2020.
156. Sardiwal, M, et al. (2017, December 7). New Targeted Attack in the Middle East by APT34, a Suspected Iranian Threat Group, Using CVE-2017-11882 Exploit. Retrieved December 20, 2017.
157. Cherepanov, A.. (2016, May 17). Operation Groundbait: Analysis of a surveillance toolkit. Retrieved May 18, 2016.
158. Kasza, A. and Reichel, D. (2017, February 27). The Gamaredon Group Toolset Evolution. Retrieved March 1, 2017.
159. Unit 42. (2022, February 3). Russia's Gamaredon aka Primitive Bear APT Group Actively Targeting Ukraine. Retrieved February 21, 2022.
160. Nicolas Verdier. (n.d.). Retrieved January 29, 2018.
161. Microsoft. (2024, September 4). Use Quick Assist to help users. Retrieved March 14, 2025.
162. Microsoft Threat Intelligence Center. (2022, February 4). ACTINIUM targets Ukrainian organizations. Retrieved February 18, 2022.
163. S2W TALON. (2022, June 16). Raccoon Stealer is Back with a New Version. Retrieved August 1, 2024.
164. Pierre Le Bourhis, Quentin Bourgue, & Sekoia TDR. (2022, June 29). Raccoon Stealer v2 - Part 2: In-depth analysis. Retrieved August 1, 2024.
165. Vrabie, V. (2021, April 23). NAIKON – Traces from a Military Cyber-Espionage Operation. Retrieved June 29, 2021.
166. Antiy CERT. (2020, April 20). Analysis of Ramsay components of Darkhotel's infiltration and isolation network. Retrieved March 24, 2021.
167. Global Threat Center, Intelligence Team. (2020, December). APT27 Turns to Ransomware. Retrieved November 12, 2021.
168. Falcone, R. (2020, July 22). OilRig Targets Middle Eastern Telecommunications Organization and Adds Novel C2 Channel with Steganography to Its Inventory. Retrieved July 28, 2020.
169. FireEye iSIGHT Intelligence. (2017, April 6). APT10 (MenuPass

55. Kohler, A. and Lopez, C. (2024, April 30). Malware: Cuckoo Behaves Like Cross Between Infostealer and Spyware. Retrieved August 20, 2024.
56. Chen, J. and Hsieh, M. (2017, November 7). REDBALDKNIGHT/BRONZE BUTLER's Daserf Backdoor Now Using Steganography. Retrieved December 27, 2017.
57. FireEye. (2018, March 16). Suspected Chinese Cyber Espionage Group (TEMP.Periscope) Targeting U.S. Engineering and Maritime Industries. Retrieved April 11, 2018.
58. FireEye. (2018, February 20). APT37 (Reaper): The Overlooked North Korean Actor. Retrieved November 17, 2024.
59. Grunzweig, J. (2018, October 01). NOKKI Almost Ties the Knot with DOGCALL: Reaper Group Uses New Malware to Deploy RAT. Retrieved November 5, 2018.
60. US-CERT. (2018, March 16). Alert (TA18-074A): Russian Government Cyber Activity Targeting Energy and Other Critical Infrastructure Sectors. Retrieved June 6, 2018.
61. Symantec Security Response. (2014, July 7). Dragonfly: Western energy sector targeted by sophisticated attack group. Retrieved September 9, 2017.
62. Slowik, J. (2021, October). THE BAFFLING BERSERK BEAR: A DECADE'S ACTIVITY TARGETING CRITICAL INFRASTRUCTURE. Retrieved December 6, 2021.
63. Mike Stokkel et al. (2024, July 18). APT41 Has Arisen From the DUST. Retrieved September 16, 2024.
64. GReAT. (2019, April 10). Gaza Cybergang Group1, operation SneakyPastes. Retrieved May 13, 2020.
65. Cybersecurity and Infrastructure Security Agency. (2020, August 26). MAR-10301706-1.v1 - North Korean Remote Access Tool: ECCENTRICBANDWAGON. Retrieved March 18, 2021.
66. Schroeder, W., Warner, J., Nelson, M. (n.d.). Github PowerShellEmpire. Retrieved April 28, 2016.
67. PwC and BAE Systems. (2017, April). Operation Cloud Hopper: Technical Annex. Retrieved April 13, 2017.
68. Department of Justice. (2018, August 01). HOW FIN7 ATTACKED AND STOLE DATA. Retrieved August 24, 2018.
69. FinFisher. (n.d.). Retrieved September 12, 2024.
70. Allievi, A.,Flori, E. (2018, March 01). FinFisher exposed: A researcher's tale of defeating traps, tricks, and complex virtual machines. Retrieved July 9, 2018.
71. Gostev, A. (2012, May 28). The Flame: Questions and Answers. Retrieved March 1, 2017.
72. Financial Security Institute. (2020, February 28). Profiling of TA505 Threat Group That Continues to Attack the Financial Sector. Retrieved July 14, 2022.
73. Patrick Wardle. (n.d.). Mac Malware of 2017. Retrieved September 21, 2018.
74. Vrabie, V. (2020, November). Dissecting a Chinese APT Targeting South Eastern Asian Government Institutions. Retrieved September 19, 2022.
75. Boutin, J. (2020, June 11). Gamaredon group grows its game. Retrieved June 16, 2020.
76. Threat Hunter Team, Symantec and Carbon Black. (2025, April 10). Shuckworm Targets Foreign Military Mission Based in Ukraine. Retrieved July 23, 2025.
77. Rusnák, Z. (2024, September 26). Cyberespionage the Gamaredon way: Analysis of toolset used to spy on Ukraine in 2022 and 2023. Retrieved October 30, 2024.
172. Splunk Threat Research Team. (2023, June 1). Do Not Cross The 'RedLine' Stealer: Detections and Analysis. Retrieved September 17, 2025.
173. Klijnsma, Y. (2018, January 23). Espionage Campaign Leverages Spear Phishing, RATs Against Turkish Defense Contractors. Retrieved November 6, 2018.
174. Zhang, X. (2024, November 8). New Campaign Uses Remcos RAT to Exploit Victims. Retrieved April 16, 2026.
175. Legezo, D. (2019, January 30). Chafer used Remexi malware to spy on Iran-based foreign diplomatic entities. Retrieved April 17, 2019.
176. Peretz, A. and Theck, E. (2021, March 5). Earth Vetala – MuddyWater Continues to Target Organizations in the Middle East. Retrieved March 18, 2021.
177. Livelli, K, et al. (2018, November 12). Operation Shaheen. Retrieved May 1, 2019.
178. Falcone, R., et al. (2018, July 27). New Threat Actor Group DarkHydrus Targets Middle East Government. Retrieved August 2, 2018.
179. Mercer, W., Rascagneres, P. (2017, April 03). Introducing ROKRAT. Retrieved May 21, 2018.
180. Mercer, W., Rascagneres, P. (2017, November 28). ROKRAT Reloaded. Retrieved May 21, 2018.
181. GReAT. (2019, May 13). ScarCruft continues to evolve, introduces Bluetooth harvester. Retrieved June 4, 2019.
182. Pantazopoulos, N.. (2018, November 8). RokRat Analysis. Retrieved May 21, 2020.
183. Jazi, Hossein. (2021, January 6). Retrohunting APT37: North Korean APT used VBA self decode technique to inject RokRat. Retrieved March 22, 2022.
184. Ray, V., Hayashi, K. (2016, February 29). New Malware 'Rover' Targets Indian Ambassador to Afghanistan. Retrieved February 29, 2016.
185. Faou, M. and Boutin, J. (2017, February). Read The Manual: A Guide to the RTM Banking Trojan. Retrieved March 9, 2017.
186. Duncan, B., Harbison, M. (2019, January 23). Russian Language Malspam Pushing Redaman Banking Malware. Retrieved June 16, 2020.
187. Cybereason Nocturnus Team. (2020, December 9). MOLERATS IN THE CLOUD: New Malware Arsenal Abuses Cloud Platforms in Middle East Espionage Campaign. Retrieved December 22, 2020.
188. Ilascu, I. (2020, December 14). Hacking group's new malware abuses Google and Facebook services. Retrieved December 28, 2020.
189. GReAT. (2017, November 1). Silence – a new Trojan attacking financial organizations. Retrieved May 24, 2019.
190. Group-IB. (2018, September). Silence: Moving Into the Darkside. Retrieved May 5, 2020.
191. Salvati, M. (2019, August 6). SILENTTRINITY Modules. Retrieved March 24, 2022.
192. BishopFox. (n.d.). Sliver Screenshot. Retrieved September 16, 2021.
193. DHS/CISA, Cyber National Mission Force. (2020, October 1). Malware Analysis Report (MAR) MAR-10303705-1.v1 – Remote Access Trojan: SLOTHFULMEDIA. Retrieved October 2, 2020.
194. FireEye. (2021, May 11). Shining a Light on DARKSIDE Ransomware Operations. Retrieved September 22, 2021.
195. FireEye. (2021, June 16). Smoking Out a DARKSIDE Affiliate's Supply Chain Software Compromise. Retrieved September 22,

81. Scott-Railton, J., et al. (2016, August 2). Group5: Syria and the Iranian Connection. Retrieved September 26, 2016.
82. Carr, N., et al. (2017, April 24). FIN7 Evolution and the Phishing LNK. Retrieved April 24, 2017.
83. Ungur, P. (n.d.). HAVOC. Retrieved August 4, 2025.
84. Shivtarkar, N. and Jain, S. (2023, February 14). Havoc Across the Cyberspace. Retrieved August 4, 2025.
85. Immersive Content Team. (2024, April 9). Havoc C2 Framework – A Defensive Operator's Guide. Retrieved August 13, 2025.
86. Knight, S.. (2020, April 16). VMware Carbon Black TAU Threat Analysis: The Evolution of Lazarus. Retrieved May 1, 2020.
87. Alexndru-Cristian Bardas. (2025, October 30). DPRK's Playbook: Kimsuky's HttpTroy and Lazarus's New BLINDINGCAN Variant. Retrieved April 8, 2026.
88. Lelli, A. (2010, January 11). Trojan.Hydraq. Retrieved February 20, 2018.
89. Falcone, R. and Lancaster, T. (2019, May 28). Emissary Panda Attacks Middle East Government Sharepoint Servers. Retrieved July 9, 2019.
90. Hromcová, Z. (2018, June 07). InvisiMole: Surprisingly equipped spyware, undercover since 2013. Retrieved July 10, 2018.
91. Hromcova, Z. and Cherpanov, A. (2020, June). INVISIMOLE: THE HIDDEN PART OF THE STORY. Retrieved July 16, 2020.
92. Brod. (2013, July 15). Signed Mac Malware Using Right-to-Left Override Trick. Retrieved July 17, 2017.
93. Thomas. (2013, July 15). New signed malware called Janicab. Retrieved July 17, 2017.
94. Unit 42. (2017, December 15). Unit 42 Playbook Viewer. Retrieved December 20, 2017.
95. Mercer, W., et al. (2017, October 22). "Cyber Conflict" Decoy Document Used in Real Cyber Conflict. Retrieved November 2, 2018.
96. Sharma, R. (2018, August 15). Revamped jRAT Uses New Anti-Parsing Techniques. Retrieved September 21, 2018.
97. Kamluk, V. & Gostev, A. (2016, February). Adwind - A Cross-Platform RAT. Retrieved April 23, 2019.
98. Yadav, A., et al. (2016, January 29). Malicious Office files dropping Kasidet and Dridex. Retrieved March 24, 2016.
99. Levene, B, et al. (2017, May 03). Kazuar: Multiplatform Espionage Backdoor with API Access. Retrieved July 17, 2018.
100. Parys, B. (2017, February 11). The KeyBoys are back in town. Retrieved June 13, 2019.
101. US-CERT. (2018, August 09). MAR-10135536-17 – North Korean Trojan: KEYMARBLE. Retrieved August 16, 2018.
102. Park, S. (2024, June 27). Kimsuky deploys TRANSLATEXT to target South Korean academia. Retrieved October 14, 2024.
103. Bermejo, L., et al. (2017, June 22). Following the Trail of BlackTech's Cyber Espionage Campaigns. Retrieved May 5, 2020.
104. Rascagneres, P. (2017, May 03). KONNI: A Malware Under The Radar For Years. Retrieved November 5, 2018.
105. Stuart Ashenbrenner, Alden Schmidt. (2024, April 25). LightSpy Malware Variant Targeting macOS. Retrieved January 3, 2025.
106. Yamout, M. (2021, November 29). WIRTE's campaign in the Middle East 'living off the land' since at least 2019. Retrieved February 1, 2022.
107. Seals, T. (2021, May 14). FIN7 Backdoor Masquerades as Ethical Hacking Tool. Retrieved February 2, 2022.
108. BI.ZONE Cyber Threats Research Team. (2021, May 13). From
198. Cybereason Nocturnus. (2022, February 1). StrifeWater RAT: Iranian APT Moses Staff Adds New Trojan to Ransomware Operations. Retrieved August 15, 2022.
199. Schlapfer, Patrick. (2022, June 6). A New Loader Gets Ready. Retrieved December 13, 2022.
200. Lunghi, D. and Lu, K. (2021, April 9). Iron Tiger APT Updates Toolkit With Evolved SysUpdate Malware. Retrieved November 12, 2021.
201. Grunzweig, J. and Miller-Osborn, J.. (2016, February 4). T9000: Advanced Modular Backdoor Uses Complex Anti-Analysis Techniques. Retrieved April 15, 2016.
202. GReAT. (2019, April 10). Project TajMahal – a sophisticated new APT framework. Retrieved October 14, 2019.
203. Cylance. (2014, December). Operation Cleaver. Retrieved September 14, 2017.
204. Lior Rochberger, Tom Fakterman, Robert Falcone. (2023, September 22). Cyberespionage Attacks Against Southeast Asian Government Linked to Stately Taurus, Aka Mustang Panda. Retrieved September 9, 2025.
205. Symantec Security Response. (2014, June 30). Dragonfly: Cyberespionage Attacks Against Energy Suppliers. Retrieved April 8, 2016.
206. Secureworks. (2019, July 24). Updated Karagany Malware Targets Energy Sector. Retrieved August 12, 2020.
207. Jiho Kim & Sebin Lee, S2W. (2024, February 7). Kimsuky disguised as a Korean company signed with a valid certificate to distribute Troll Stealer (English ver.). Retrieved January 17, 2025.
208. Symantec Threat Hunter Team. (2024, May 16). Springtail: New Linux Backdoor Added to Toolkit. Retrieved January 17, 2025.
209. Adam Burgher. (2021, June 10). BackdoorDiplomacy: Upgrading from Quarian to Turian. Retrieved September 1, 2021
210. O'Leary, J., et al. (2017, September 20). Insights into Iranian Cyber Espionage: APT33 Targets Aerospace and Energy Sectors and has Ties to Destructive Malware. Retrieved February 15, 2018.
211. Matsuda, A., Muhammad I. (2018, September 13). APT10 Targeting Japanese Corporations Using Updated TTPs. Retrieved September 17, 2018.
212. Hiroaki, H. (2025, April 30). Earth Kasha Updates TTPs in Latest Campaign Targeting Taiwan and Japan. Retrieved April 17, 2026.
213. Hiroaki, H. (2024, November 26). Guess Who's Back - The Return of ANEL in the Recent Earth Kasha Spear-phishing Campaign in 2024. Retrieved April 17, 2026.
214. Dominik Breitenbacher. (2025, March 18). Operation AkaiRyū: MirrorFace invites Europe to Expo 2025 and revives ANEL backdoor. Retrieved May 22, 2025.
215. Caragay, R. (2015, March 26). URSNIF: The Multifaceted Malware. Retrieved June 5, 2019.
216. Sioting, S. (2013, June 15). BKDR_URSNIF.SM. Retrieved June 5, 2019.
217. Salem, E. et al. (2020, May 28). VALAK: MORE THAN MEETS THE EYE . Retrieved June 19, 2020.
218. Lancaster, T., Cortes, J. (2018, January 29). VERMIN: Quasar RAT and Custom Malware Used In Ukraine. Retrieved July 5, 2018.
219. FBI. (2026, March 20). FBI Flash: FLASH-20260320-001:Government of Iran Cyber Actors Deploy Telegram C2 to Push Malware to Identified Targets. Retrieved April 20, 2026.
220. CISA et al.. (2024, February 7). PRC State-Sponsored Actors Compromise and Maintain Persistent Access to U.S. Critical Infrastructure. Retrieved May 15, 2024.

111. ITOCHU. (2024, January 24). The Endless Struggle Against APT10: Insights from LODEINFO v0.6.6 - v0.7.3 Analysis. Retrieved April 17, 2026.
112. Raggi, M. Schwarz, D.. (2019, August 1). LookBack Malware Targets the United States Utilities Sector with Phishing Attacks Impersonating Engineering Licensing Boards. Retrieved February 25, 2021.
113. Cybereaon Security Services Team. (n.d.). Your Data Is Under New Lummanagement: The Rise of LummaStealer. Retrieved March 22, 2025.
114. Jurčacko, F. (2024, May 15). To the Moon and back(doors): Lunar landing in diplomatic missions. Retrieved June 26, 2024.
115. ESET. (2019, July). MACHETE JUST GOT SHARPER Venezuelan government institutions under attack. Retrieved September 13, 2019.
116. Kaspersky Global Research and Analysis Team. (2014, August 20). El Machete. Retrieved September 13, 2019.
117. The Cylance Threat Research Team. (2017, March 22). El Machete's Malware Attacks Cut Through LATAM. Retrieved September 13, 2019.
223. Mac Threat Response, Mobile Research Team. (2020, August 13). The XCSSET Malware: Inserts Malicious Code Into Xcode Projects, Performs UXSS Backdoor Planting in Safari, and Leverages Two Zero-day Exploits. Retrieved October 5, 2021.
224. Nart Villeneuve, Randi Eitzman, Sandor Nemes & Tyler Dean, Google Cloud. (2017, October 5). Significant FormBook Distribution Campaigns Impacting the U.S. and South Korea. Retrieved March 11, 2025.
225. Gustavo Palazolo, Netskope. (2022, March 11). New Formbook Campaign Delivered Through Phishing Emails. Retrieved March 11, 2025.
226. Schwarz, D., Sopko J. (2018, March 08). Donot Team Leverages New Modular Malware Framework in South Asia. Retrieved June 11, 2018.
227. ESET. (2018, November 20). Sednit: What's going on with Zebrocy?. Retrieved February 12, 2019.
228. Lee, B., Falcone, R. (2018, December 12). Dear Joohn: The Sofacy Group's Global Campaign. Retrieved April 19, 2019.
229. ESET Research. (2019, May 22). A journey to Zebrocy land. Retrieved June 20, 2019.
230. Accenture Security. (2018, November 29). SNAKEMACKEREL. Retrieved April 15, 2019.
231. CISA. (2020, October 29). Malware Analysis Report (AR20-303B). Retrieved December 9, 2020.
232. Ebach, L. (2017, June 22). Analysis Results of Zeus.Variant.Panda. Retrieved November 5, 2018.
233. Gross, J. (2016, February 23). Operation Dust Storm. Retrieved December 22, 2021.
234. Fraser, N., et al. (2019, August 7). Double DragonAPT41, a dual espionage and cyber crime operation APT41. Retrieved September 23, 2019.